



HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which region of the small intestine is the longest?

EASY

- A. Duodenum
- B. Jejunum
- C. Ileum
- D. All three are equal in length
- E. The duodenum and jejunum together exceed the ileum

2. Vitamin B12 absorption requires intrinsic factor and occurs specifically in which region?

EASY

- A. Duodenum
- B. Jejunum
- C. Ileum
- D. Throughout the entire small intestine equally
- E. Stomach

3. Which monosaccharide is absorbed by facilitated diffusion rather than sodium-dependent co-transport?

EASY

- A. Glucose
- B. Galactose
- C. Fructose
- D. Sucrose
- E. Maltose

4. Where are brush border enzymes located?

EASY

- A. Secreted freely into the intestinal lumen
- B. Bound to the microvillus membrane of enterocytes
- C. Produced by the pancreas
- D. Found only in gastric glands
- E. Secreted by Brunner's glands

5. A patient undergoes ileal resection for Crohn's disease. Several years later, they develop macrocytic anemia. Which deficiency is most directly explained by the surgery?

MEDIUM

- A. Iron deficiency
- B. Vitamin B12 deficiency
- C. Calcium deficiency
- D. Folate deficiency from the jejunum
- E. Vitamin K deficiency

6. A patient with a lymphatic obstruction (e.g., intestinal lymphangiectasia) has malabsorption specifically of long-chain fats, while glucose and amino acid absorption remain intact. What explains this selective pattern?

MEDIUM

- A. Long-chain fatty acids and monoglycerides are packaged into chylomicrons and absorbed via the lymphatic lacteals, unlike glucose and amino acids which go to the portal blood
- B. All nutrients use the same lymphatic pathway, so this pattern is implausible
- C. Fat absorption depends entirely on the portal vein, so lymphatic obstruction should not affect it
- D. Glucose and amino acids also require lymphatic transport
- E. This pattern indicates pancreatic insufficiency, not a lymphatic problem

7. A patient with lactase deficiency develops bloating and diarrhea after consuming dairy. At which specific step of carbohydrate handling is the defect located?

MEDIUM

- A. Salivary amylase action on starch
- B. Pancreatic amylase action on starch
- C. Brush border hydrolysis of the disaccharide lactose into monosaccharides
- D. Sodium-dependent co-transport of glucose into the enterocyte
- E. Chylomicron formation

8. A patient with vitamin D deficiency is found to have low calcium absorption despite an intact duodenum. Which mechanism best explains this link?

MEDIUM

- A. Calcium absorption in the duodenum is vitamin D-dependent
- B. Calcium absorption occurs only in the ileum and is unrelated to vitamin D
- C. Vitamin D is required for iron absorption, not calcium
- D. Calcium absorption depends on intrinsic factor, like B12
- E. Vitamin D deficiency has no bearing on mineral absorption

9. A patient with chronic pancreatic insufficiency has malabsorption of protein and starch but relatively preserved absorption of small peptides via PepT1. Which explanation best accounts for the residual peptide absorption despite reduced pancreatic enzyme output?

MEDIUM

- A. Brush border peptidases and the PepT1 transporter can still process short peptides that reach the intestine, even with reduced pancreatic proteolysis, though overall digestion is diminished
- B. PepT1 requires pancreatic enzymes directly to function
- C. Peptide absorption is entirely independent of any digestive step
- D. Pancreatic insufficiency eliminates all protein digestion capacity, so this scenario is implausible
- E. PepT1 only transports intact triglycerides

10. Which small intestinal region is correctly matched with its most distinguishing structural feature?

MEDIUM

- A. Duodenum — most abundant villi and circular folds
- B. Jejunum — richly vascular with the most prominent circular folds and villi
- C. Ileum — Brunner's glands secreting alkaline mucus
- D. Ileum — the shortest of the three segments
- E. Duodenum — contains Peyer's patches

11. A patient with celiac disease has villous atrophy predominantly affecting the proximal small intestine. Which functional consequence would be expected earliest, given the region most affected?

MEDIUM

- A. Isolated vitamin B12 deficiency, since celiac disease selectively destroys ileal mucosa
- B. Impaired absorption of iron, calcium, and macronutrients normally handled by the duodenum/jejunum, before ileal-dependent B12 absorption is affected
- C. No absorptive consequences, since the ileum compensates completely
- D. Selective loss of only water absorption
- E. Only fat-soluble vitamin absorption is affected, sparing carbohydrates and protein

12. A patient has surgical resection of the duodenum only, with jejunum and ileum intact and re-anastomosed to the stomach. Which secretory function would be most directly lost?

MEDIUM

- A. Secretin and CCK release, and Brunner's gland alkaline mucus secretion
- B. Vitamin B12 absorption
- C. Bile salt reabsorption
- D. Peyer's patch immune surveillance
- E. Chylomicron formation entirely

13. A patient with short bowel syndrome after extensive jejunal resection, but with duodenum and ileum both intact, is expected to have the greatest functional impact on which process?

MEDIUM

- A. Vitamin B12 absorption, since the jejunum is essential for this
- B. Bulk absorption of carbohydrates, proteins, and lipids, since the jejunum is the main site of macronutrient absorption despite duodenum/ileum being spared
- C. Bile salt reabsorption, since this occurs mainly in the jejunum
- D. Gastric acid neutralization, since this depends on the jejunum
- E. No functional impact, since the ileum can fully substitute for the jejunum in all absorptive functions

14. A patient's stool osmotic gap testing suggests unabsorbed solute is drawing water into the lumen after a large meal rich in a specific sugar substitute known to be poorly absorbed. Which absorption mechanism, if defective or simply absent for that sugar, best explains this pattern?

MEDIUM

- A. Passive paracellular absorption occurring efficiently for all solutes regardless of any transporter
- B. A specific carrier-mediated or co-transport mechanism (as with glucose/galactose via SGLT1, or fructose via GLUT5) being absent for that particular sugar, leaving it osmotically active in the lumen
- C. All sugars are absorbed identically via simple diffusion
- D. The sugar must be a lipid, not a carbohydrate, to cause this effect
- E. This pattern can only be explained by pancreatic disease

15. A patient has combined loss of ileal function (post-resection) and reduced bile acid synthesis. Predict the compounded effect on fat absorption, considering that bile salts are normally reabsorbed in the ileum for enterohepatic recycling.

HARD

- A. No compounded effect, since bile acid synthesis and ileal absorption are entirely independent processes
- B. A more severe fat malabsorption than either defect alone would predict, because ileal resection also interrupts enterohepatic recycling of bile salts, further reducing the already-limited bile salt pool available for micelle formation
- C. Improved fat absorption, since ileal resection increases bile salt delivery to the jejunum
- D. This combination would only affect protein absorption, not fat
- E. Bile salts are irrelevant to fat absorption; only pancreatic lipase matters

16. A physiologist notes that glucose and amino acid absorption both use sodium-dependent co-transport, while fructose uses facilitated diffusion. In a patient with a mutation causing complete loss of the Na⁺/K⁺-ATPase in enterocyte basolateral membranes, which absorptive process would be most directly and immediately impaired?

HARD

- A. Only fructose absorption, since it depends most directly on ATP
- B. Glucose and amino acid absorption, since sodium-dependent co-transport relies on the low intracellular Na⁺ gradient maintained by the Na⁺/K⁺-ATPase; fructose absorption (facilitated diffusion, sodium-independent) would be relatively less directly dependent on this specific pump
- C. Neither process would be affected, since both are entirely passive
- D. Only lipid absorption would be affected
- E. All absorption would immediately stop, including passive diffusion of water

17. A patient has a rare congenital defect that selectively prevents PepT1 (the H⁺-coupled di/tripeptide transporter) from functioning, while all other transport systems (amino acid transporters, brush border peptidases) are normal. Predict the most likely outcome for protein absorption.

HARD

- A. Complete failure of all protein absorption, since PepT1 is the only route for amino acid-derived nutrients
- B. Largely preserved protein absorption, since brush border peptidases can still hydrolyze peptides to free amino acids for absorption via sodium-dependent amino acid transporters, providing a parallel route
- C. Selective failure of only fat absorption
- D. No effect whatsoever on any nutrient absorption
- E. Complete failure of carbohydrate absorption as a secondary consequence

18. A comparative physiology question asks why villi height and density are greatest in the jejunum, moderate in the duodenum, and lowest in the ileum, despite all three segments being part of the "absorptive" small intestine. Which explanation best fits the described anatomy and function?

HARD

- A. Villus density has no relationship to absorptive workload; it is purely random anatomical variation
- B. The jejunum bears the greatest bulk macronutrient absorptive load (carbohydrates, protein, lipids) and is structurally optimized with the densest villi/folds for this task, while the ileum, handling more specialized, lower-volume absorption (B12, bile salts) and immune surveillance (Peyer's patches), has comparatively less villous density
- C. The duodenum should have the highest villus density because it receives chyme first
- D. Villus density decreases distally purely due to acid damage accumulating along the tract
- E. The ileum has the highest villus density of the three segments

19. A patient with short bowel syndrome after resection of both jejunum and a large portion of ileum, but with an intact duodenum, develops multiple deficiencies. Which combination of findings would be the most specific and direct consequence of losing the ileal segment specifically, as opposed to a general consequence of reduced total absorptive surface?

HARD

- A. Generalized malnutrition and weight loss only, which are non-specific and could result from loss of any segment
- B. Vitamin B12 deficiency and bile salt malabsorption (with resulting fat malabsorption from a depleted bile salt pool), since these are the ileum-specific functions
- C. Isolated calcium deficiency, since calcium absorption is ileum-specific
- D. Isolated iron deficiency, since iron absorption is ileum-specific
- E. No specific deficiency, since the duodenum can fully compensate for all ileal functions

20. A drug is designed to selectively block the SGLT1 (sodium-glucose co-transporter) in the small intestine to treat a metabolic condition. Predict the effect on fructose absorption and explain the physiological basis.

HARD

- A. Fructose absorption would also be blocked, since it uses the same transporter as glucose
- B. Fructose absorption would be relatively preserved, since fructose is absorbed via GLUT5-mediated facilitated diffusion, a distinct, sodium-independent pathway from SGLT1
- C. Fructose absorption would increase dramatically to compensate
- D. Fructose absorption depends entirely on SGLT1 despite using a different name for the same protein
- E. This drug would have no measurable effect on any monosaccharide absorption

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	The ileum comprises about 3/5 of small intestinal length, making it the longest segment.
2	C	EASY	Vitamin B12 (with intrinsic factor) is absorbed specifically in the ileum.
3	C	EASY	Fructose is absorbed via facilitated diffusion, unlike glucose and galactose which use sodium co-transport.
4	B	EASY	Brush border enzymes are membrane-bound on the microvilli, not secreted into the lumen.
5	B	MEDIUM	B12 absorption is site-specific to the ileum; its loss causes B12 deficiency and macrocytic (megaloblastic) anemia.
6	A	MEDIUM	Long-chain fatty acids/monoglycerides are re-esterified to triglycerides, packaged into chylomicrons, and transported via lacteals into lymph — the one macronutrient pathway that depends on an intact lymphatic system, unlike glucose/amino acids which go directly to portal blood.

7	C	MEDIUM	Lactase is a brush border enzyme; its deficiency blocks the final hydrolysis of lactose into glucose and galactose, leaving undigested lactose to cause osmotic diarrhea and bacterial fermentation (bloating).
8	A	MEDIUM	Duodenal calcium absorption is vitamin D-dependent; deficiency directly reduces calcium uptake.
9	A	MEDIUM	While pancreatic proteases (trypsin, chymotrypsin) are major contributors to protein digestion, brush border peptidases and the H ⁺ -coupled PepT1 transporter provide some redundancy for absorbing small peptides, though overall digestion and absorption remain impaired with significant pancreatic insufficiency.
10	B	MEDIUM	The jejunum is richly vascular with the most prominent plicae circulares and villi; Brunner's glands and shortest length belong to the duodenum, and Peyer's patches belong to the ileum.
11	B	MEDIUM	Celiac disease characteristically affects the proximal small bowel (duodenum/proximal jejunum) most severely, so iron, calcium, and general macronutrient absorption (jejunal-dependent) are affected earlier/ more severely than B12, which depends on the ileum.
12	A	MEDIUM	The duodenum secretes secretin and CCK and contains Brunner's glands producing alkaline mucus to neutralize gastric acid; these functions would be lost with duodenal resection, while B12 absorption and Peyer's patches (ileum) would be preserved.
13	B	MEDIUM	Even with duodenum and ileum spared, losing the jejunum — the primary site of carbohydrate, protein, and lipid absorption with the most villi and vascularity — has a major impact on overall macronutrient absorption.
14	B	MEDIUM	Sugars without a dedicated transport mechanism in the small intestine (e.g., certain sugar alcohols) remain osmotically active in the lumen, drawing water in and causing osmotic diarrhea — illustrating why specific transporters (SGLT1 for glucose/galactose, GLUT5 for fructose) matter for absorption.
15	B	HARD	Ileal resection removes the site of bile salt reabsorption (enterohepatic circulation), so a reduced bile acid pool cannot be replenished efficiently, compounding fat malabsorption beyond either defect (reduced synthesis or lost reabsorption) alone.
16	B	HARD	Na ⁺ /K ⁺ -ATPase maintains the low intracellular sodium gradient that drives all sodium-dependent co-transport (glucose, galactose, amino acids); fructose's facilitated diffusion via GLUT5 does not depend on this sodium gradient, making it relatively more independent of Na ⁺ /K ⁺ -ATPase function, at least in the short term.
17	B	HARD	Protein absorption has two parallel routes: free amino acids via sodium-dependent transporters, and di/tripeptides via PepT1 (after which they are hydrolyzed intracellularly). Loss of PepT1 alone would not abolish protein absorption because brush border peptidases can still generate free amino acids for the intact amino acid transport pathway.
18	B	HARD	Structure follows function: the jejunum, handling the bulk of macronutrient absorption, has the most prominent circular folds and villi, while the ileum — specialized for lower-volume, targeted absorption (B12, bile salts) and immune surveillance via Peyer's patches — has comparatively fewer folds/villi.
19	B	HARD	B12 absorption and bile salt reabsorption (enterohepatic circulation) are ileum-specific functions with no compensation from the duodenum; their loss (with resulting fat malabsorption from bile salt depletion) is the most specific, site-attributable consequence, distinct from generalized malabsorption from reduced total surface area.
20	B	HARD	Fructose is absorbed via facilitated diffusion (a distinct sodium-independent mechanism, commonly attributed to GLUT5), unlike glucose and galactose which require SGLT1; selectively blocking SGLT1 would spare fructose absorption while blocking glucose/galactose uptake.